

Marie Kasongo

Boston, MA | (857) 300-3902 | marie_kasongo@brown.edu | [linkedin.com/in/marie-kasongo/](https://www.linkedin.com/in/marie-kasongo/)

EDUCATION

Brown University
Bachelors of Science, Biomedical Engineering
Activities & Organizations: Bonner Community Fellow (for Social Justice Work); Vice President of PR & Marketing of Alpha Chi Omega; Event Coordinator of African Students Association; Meiklejohn Peer Advisor; National Society of Black Engineers; Valedictorian Project Scholar; Questbridge Scholar
Courses: Fluid Mechanics, Engineering Design, Thermodynamics, Signal and Circuits, Viral Epidemics, Differential Equations

Providence, RI
Anticipated May 2028

RESEARCH EXPERIENCE

Tripathi BioEngineering Lab, Assay Development Team
Research Assistant
November 2025 – Present

- Develop HepG2 3D-spheroid cell culture models to study how spheroid structure and growth conditions affect tissue dissociation
- Design and optimize enzymatic and electrical dissociation protocols to generate single-cell suspensions from spheroids
- Determine minimum and maximum cell processing ranges for electrical tissue dissociation chambers and evaluate whether electrical dissociation improves recovery from aged spheroids

Panoplia Laboratories
Research Technician
June – August 2025

- Conducted wet-lab research in a 13-week study evaluating AI-assisted lab training, completing protocol-based benchmarks under defined accuracy criteria
- Performed mammalian cell culture and AAV production in HEK293T cells; executed DNA fragment cloning, plasmid preparation, and construct validation; assisted with qPCR workflows for construct verification
- Advanced through study phases by rapidly mastering new protocols while maintaining organized digital lab documentation, safety compliance, and collaboration with senior researchers

Wong Bioimaging Engineering Lab - Brown University
Research Assistant
December 2024-May 2025

- Investigated correlation between morphological anomalies and circulating tumor cell (CTC) development under varying microenvironment stiffness, under the mentorship of Dr. Ian Wong
- Cultured CTCs and performed cytoplasmic (CellMask, Cy5) and nuclear (Hoechst, DAPI) staining for fluorescent imaging
- Captured high-resolution images and developed optimized Cellpose segmentation pipelines with manual boundary correction to detect morphological anomalies
- Applied PCA and qualitative visualization methods (bar graphs, T-charts) to classify morphological traits and determine how microenvironment stiffness influences CTC behavior and metastatic potential

PROJECTS

Spring-Based Mass Launchers, School of Engineering
Spring 2025

- Simulated a vertical spring-powered mechanical system in MATLAB (ODE45) to predict launch velocity
- Applied force balance and mass-spring dynamics to model motion and optimize design parameters
- Implemented event-detection logic to identify equilibrium return and refine model accuracy

Hip Replacement Joint, School of Engineering
Fall 2024

- Designed and analyzed a hip replacement joint in CAD, performing stress-strain testing to validate performance under 600N loads
- Developed 50+ hip resurfacing concepts in a team, incorporating material selection and market-informed design constraints

COMMUNITY ENGAGEMENT

Connect for Health, Hashboro Hospital
Patient Resource Advocate
Fall 2025- Present

- Worked 1:1 with low-income pediatric patients and families over 3–12 month case periods to coordinate long-term resource navigation and continuity-of-care support
- Assisted clients in accessing utilities assistance, housing/shelter support, transportation affordability programs, furniture resources, clothing drives, public benefits, and other social determinant of health services
- Connected patients and families with appropriate medical, legal, mental health, physical health, and social work resources while also supporting front-desk patient assistance and walk-in family needs within a pediatric hospital setting

DownCity Design
Youth Design Mentor
Spring 2025

- Mentored high school students in human-centred design process to build a portable garden bed with integrated seating for a local Providence elderly residential community; incorporated residents feedback into revisions
- Led student course development, project planning, and workflow coordination (4–6 hrs/week)

SKILLS & PUBLICATIONS

Technical: Fluorescence imaging and segmentation (Cellpose & Cellprofiler), qPCR assays execution, mammalian cell culture

Software: Python, R, MATLAB, CAD/Fusion 360, and Microsoft Office 360

Certifications: EHR (Epic Systems) Management, HIPAA, Synthetic Biology Experience, AI prompt Engineering

Languages: Fluent French and Beginner Spanish

Publications: “The Beauty Of Mindful Living: How To Create Rituals Instead Of Routines” (2025, HerCampus) and “7 Free STEM Summer Programs For Students” (2023, Channel Kindness)